SVKM'S
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Mukesh Patel School of Technology Management & Engineering

Project registration form

Department: Artificial Intelligence
Program: B.Tech
Semester: VII

Project Team details

S No	SAP ID	Roll No	Name	Signature
1	70122300012	I023	Yash Garg	
2	70122300051	I054	Anumay Pandey	
3	70122300039	I037	Yash Kothari	

Three broad domains in which the project team intends to work

S No	Domain	Project idea in brief
1	Speech Signal Processing & Audio Forensics	Detecting synthetic speech and generative voice clones by identifying vocoder phase inconsistencies and spectral artifacts.
2	Self-Supervised Learning & Deep Learning	
3	Biometrics & Cybersecurity	Designing an automated two-stage authentication framework coupling deepfake detection with biometric speaker verification.

A.2 Topic approval form

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Type of Project (Tick any one)

Type of project	
Application	
Product	
Research	.

Project details

Project Title	**Bilingual Speech Detector**
Project Objectives	1. Build a high-accuracy, cross-lingual audio anti-spoofing system for English and Hindi voice clones. 2. Exploit multi-layer representations from frozen XLS-R 300M using Selective Layer Summarization (SLS). 3. Eliminate shortcut learning on non-speech cues via silence-invariant augmentation and onset sliding windows. 4. Implement an end-to-end biometric verification pipeline with ECAPA-TDNN and automated interlock gating.
Domain	Audio Forensics, Self-Supervised Speech Processing, and Biometrics

Motivation	Generative voice cloning (XTTS, FreeVC) and commercial TTS threaten voice-biometric security. Existing detectors are largely English-trained and fail on Indian languages like Hindi, while overfitting to silence shortcuts and partial splices. This project creates a robust, bilingual, silence-invariant forensic system coupling deepfake detection with biometric authentication.
Expected outcomes	? Robust bilingual deepfake detector (EER < 3%) across diverse English and Hindi voice clones. ? Elimination of silence shortcuts, sliding-window partial-spoof detection, and biometric interlock.

Latest References

S No	Publication (Author, "Title", Journal/Book/Conference, Date, Page, Volume, year)
1	E. A. AlBadawy, A. Gibiansky, Q. He, J. Wu, M.-C. Chang, and S. Lyu, ?VocBench: A Neural Vocoder Benchmark for Speech Synthesis,? in Proc. IEEE ICASSP, 2022. arXiv:2112.03099.
2	J. Yamagishi et al., ?ASVspoof 2021: Accelerating progress in spoofed and deepfake speech detection,? in Proc. ASVspoof Workshop, 2021.
3	J. Yi, C. Wang, J. Tao, X. Zhang, C. Y. Zhang, and Y. Zhao, ?Audio Deepfake Detection: A Survey,? arXiv:2308.14970, 2023. Available: https://arxiv.org/abs/2308.14970.
4	Q. Zhang, S. Wen, and T. Hu, ?Audio Deepfake Detection with Self-Supervised XLS-R and SLS Classifier,? in Proc. 32nd ACM International Conference on Multimedia, pp. 6765?6773, 2024.
5	A. Babu, C. Wang, A. Tjandra, et al., ?XLS-R: Self-supervised Crosslingual Speech Representation Learning at Scale,? in Proc. Interspeech, 2022.

Status of the project after Review 1

Comments/Remarks	
Name and Signature of the guide	
Name and signature of panel members for review 1	
Signature of HoD	

If any change in project title after review 1 presentation as per comments received from panel and areas of concern to be noted here